

A.J. SADLER

MATHEMATICS APPLICATIONS

UNIT

3



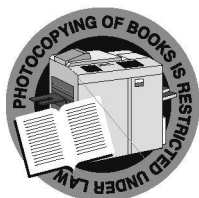
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Preface.

This text targets Unit Three of the West Australian course *Mathematics Applications* a course that is organised into four units altogether, the first two for year eleven and the last two for year twelve.

UNIT
ONE

UNIT
TWO

UNIT
THREE

UNIT
FOUR

This West Australian course, *Mathematics Applications*, is based on the Australian National Curriculum Senior Secondary course *General Mathematics*. At the time of writing there are only small differences between the Unit Three content of these two courses, the main difference being that specific examples mentioned in the National syllabus are moved to the glossary of the West Australian syllabus. Hence this text is also suitable for anyone following Unit Three of the National Curriculum course, *General Mathematics*.

The book contains text, examples and exercises containing many carefully graded questions. A student who studies the appropriate text and relevant examples should make good progress with the exercise that follows.

The book commences with a section entitled *Preliminary work*. This section briefly outlines work of particular relevance to this unit that students should either already have some familiarity with from the mathematics studied in earlier years, or for which the brief outline included in the section may be sufficient to bring the understanding of the concept up to the necessary level.

As students progress through the book they will encounter questions involving this preliminary work in the *Miscellaneous Exercises* that feature at the end of each chapter. These miscellaneous exercises also include questions involving work from preceding chapters to encourage the continual revision needed throughout the unit.

Some chapters commence with, or contain, a "Situation" or two for students to consider, either individually or as a group. In this way students are encouraged to think and discuss a situation, which they are able to tackle using their existing knowledge, but which acts as a fore-runner and stimulus for the ideas that follow. Students should be encouraged to discuss their solutions and answers to these situations and perhaps to present their method of solution to others. For this reason answers to these situations are generally not included in the book.

At times in this series of books I have found it appropriate to go a little beyond the confines of the syllabus for the unit involved. In this regard, in the chapter introducing recursively defined sequences I sometimes go beyond the types specifically mentioned in the syllabus, when considering networks I include use of the word traversable and the idea of a tree and when considering linear regression I include mention of the " x on y " regression line. I leave it up to the readers and teachers to decide whether to cover these aspects or not.

Alan Sadler

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Important note.

This series of texts has been written based on my interpretation of the appropriate *Mathematics Applications* syllabus documents as they stand at the time of writing. It is likely that as time progresses some points of interpretation will become clarified and perhaps even some changes could be made to the original syllabus. I urge teachers of the *Mathematics Applications* course, and students following the course, to check with the appropriate curriculum authority to make themselves aware of the latest version of the syllabus current at the time they are studying the course.

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Alan Sadler